

Amal Thomas

Full Stack Developer | MERN Stack Developer

amalthomaschennattu@gmail.com | +91 9895114935 | Alappuzha, Kerala, India

amalthomas-portfolio.vercel.app | linkedin.com/in/amalthomas26 | github.com/amalthomas26

Professional Summary

Fresher Full Stack Developer building production-grade MERN applications with hands-on experience in real-time chat, WebRTC audio/video calling, authentication, Docker, AWS, Cloudflare, and CI/CD. Focused on clean architecture, reliability, performance, and deployment-ready frontend-backend systems.

Technical Skills

Frontend: React 19, TypeScript, JavaScript ES6, Vite, Tailwind CSS, Zustand, Redux, HTML5, CSS3

Backend: Node.js, Express.js, REST APIs, GraphQL, Socket.IO, WebRTC

Database: MongoDB

DevOps: Docker, AWS EC2, Nginx, Cloudflare, GitHub Actions, CI/CD

Auth & Security: JWT, Google OAuth, 2FA, OTP, RBAC, Helmet, Rate Limiting

Concepts: Real-Time Systems, Backend Architecture, Full Stack Development

Project Experience

BlinkChat - Full-Stack Real-Time Chat Application | Mar 2026 - Jun 2026 (First Week) | Node.js, Express, MongoDB, Socket.IO, WebRTC, React, **TypeScript**, **Docker**, AWS, **Nginx**, Cloudflare | Live: <https://blink-chat.app/>

- Engineered a production **WebRTC** video/audio calling system with a self-hosted **Coturn TURN server** on **AWS EC2** using HMAC-SHA1 short-lived credentials.
- Designed a 9-state call lifecycle finite state machine with 15-second mid-call reconnect recovery and adaptive resolution fallback from 1080p to 720p to audio.
- Built a typed **Socket.IO v4** real-time layer with **JWT** middleware, connection state recovery, typing indicators, online presence tracking, and per-room guards.
- Architected a 3-container **Docker** Compose deployment behind **Cloudflare CDN** on AWS EC2 with **GitHub Actions CI/CD** and **Nginx** for zero-touch deploys.
- Implemented **JWT auth**, **Google OAuth**, **2FA**, OTP email flow, Helmet, and per-route rate limiting.
- Built messaging features including reactions, replies with deletion-safe snapshots, forwarding, audio recording, group management, and cursor-based search.

PrimePicks - E-commerce Web Application | Jan 2026 - Feb 2026 | **React**, **Vite**, JavaScript, CSS, AWS Elastic Beanstalk, GitHub Actions

- Built a responsive e-commerce app with product listing, product detail pages, cart management, and **CI/CD**-based deployment.

GraphQL User Management System - Node.js, Express, GraphQL, MongoDB, JWT, RBAC | Feb 2026

- Built a GraphQL API with **JWT authentication**, **RBAC authorization**, full CRUD, and protection against over-fetching.

Education

Higher Secondary Certificate - Computer Science, Leo XIII HSS Alappuzha | 2019 - 2021 | Grade A1

Key Strengths

Quick Learner | Problem Solving | Backend Architecture | Real-Time Systems | Team Collaboration | Production Deployments | Clean Code